

L4 Safety Island — the four declared behaviours

The island’s node graph (phase 5 of the series), lit once per behaviour of §3.1

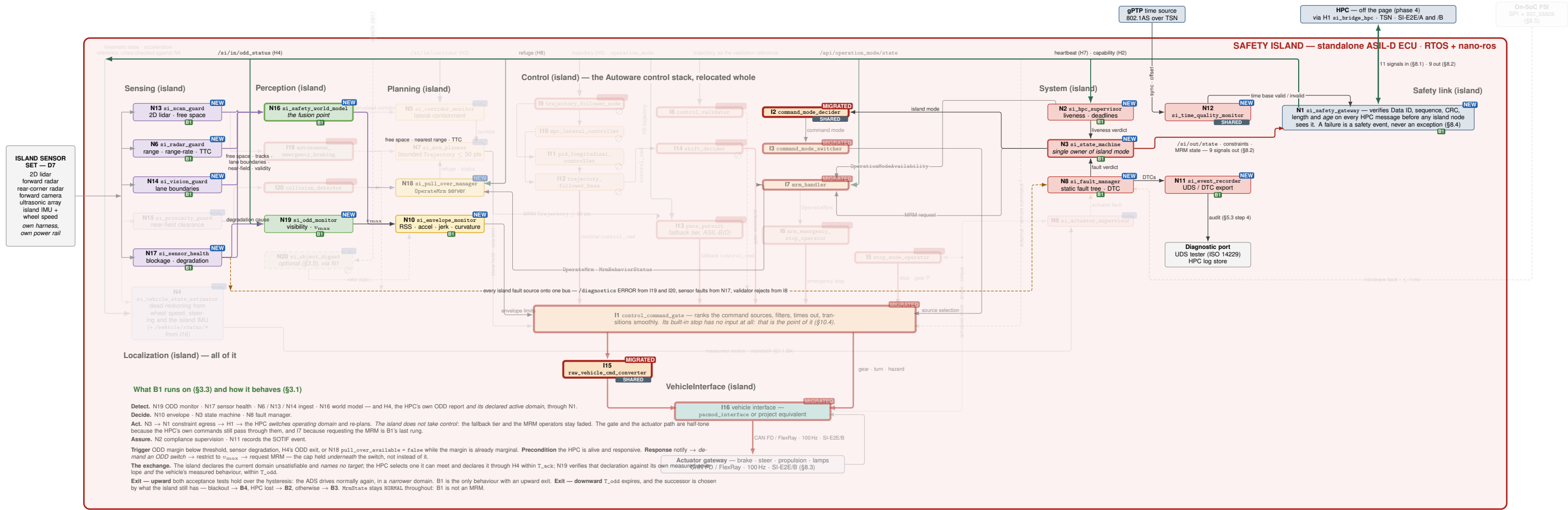
B1	Degrade the ADS when the ODD cannot be held	The island constrains and demands an ODD switch; the HPC executes. The only behaviour with an upward exit
B2	Keep the vehicle in its lane after loss of the HPC	The island takes authority and holds the lane for a bounded window while the lane is still observable
B3	Pull over if the ODD continues to fail	A placed stop in a verified refuge, nominated by the HPC or found by the island
B4	Emergency stop when the island is blind	An unplaced stop with no exteroceptive sensor at all: the island brakes and declines to steer

On every page the drawing is the same. What the behaviour runs on — the nodes named in its §3.3 row — and the SHARED nodes every behaviour runs through are in **full colour**; what it only passes through by §3.1 is **half-tone**; the rest of the island is **faded**. An edge is drawn at the weaker of its two ends, and every lit node keeps at least one lit edge in and one out. The lower-left panel of each page lists the behaviour’s Detect · Decide · Act · Assure nodes and its trigger, precondition, response and exit.

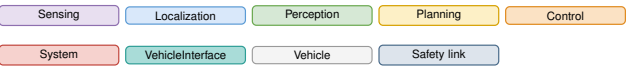
Derived from L4SafetyIsland_SoftwareDesign.md, version 20260918, without modifying it.

Phase 5 · B1 — Degrade the ADS when the ODD cannot be held

When the operating envelope is no longer satisfied, require the ADS to move to an operating domain it *can* satisfy — progressively constraining it, and demanding that it constrain itself, until it does — before any authority is taken. The phase-5 drawing, unchanged: what B1 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



Stacks — the colours of ArchitectureMain.pdf

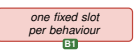


Where each node came from



18 migrated + 19 new = 37, plus the optional N20. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the external / remote command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table.

Behaviours each node enables (§3.3)



B1 degrade the ADS · B2 lane-keep hold after HPC loss · B3 pull over · B4 blind emergency stop. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

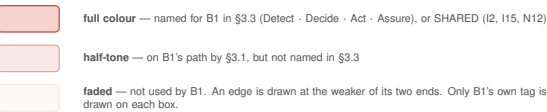
Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9-I12 and I14 track the HPC's nominal trajectory, which B2-B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages



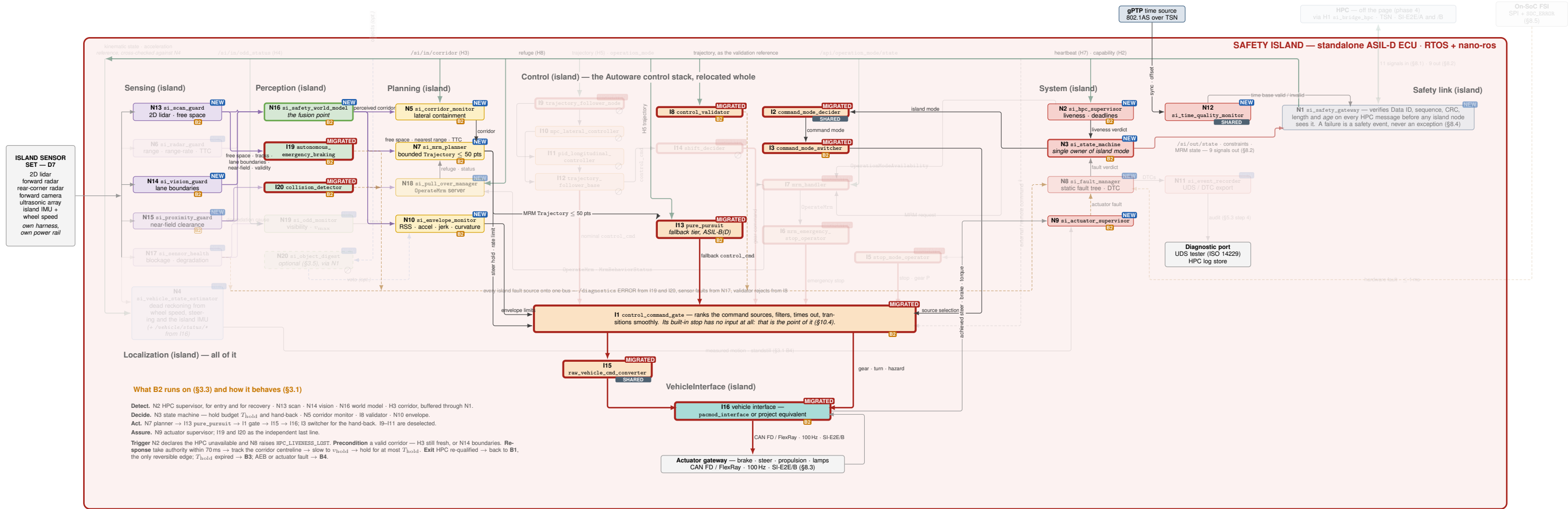
In this view — B1, degrade the ADS



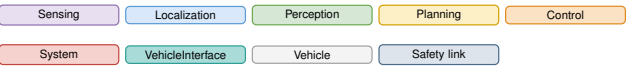
Every edge on this page, with the design-document section it is taken from, is catalogued in ../messageflow/, which draws the same 37 nodes coloured by execution partition instead.

Phase 5 · B2 — Keep the vehicle in its lane after loss of the HPC

When the HPC is unavailable and B1 therefore cannot be used, take authority and hold the vehicle in its current lane — for a bounded window — while the lane is still observable.
The phase-5 drawing, unchanged: what B2 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



Stacks — the colours of ArchitectureMain.pdf

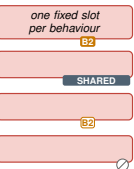


Where each node came from



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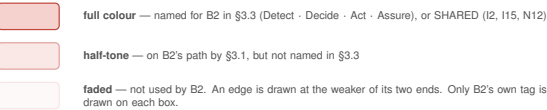
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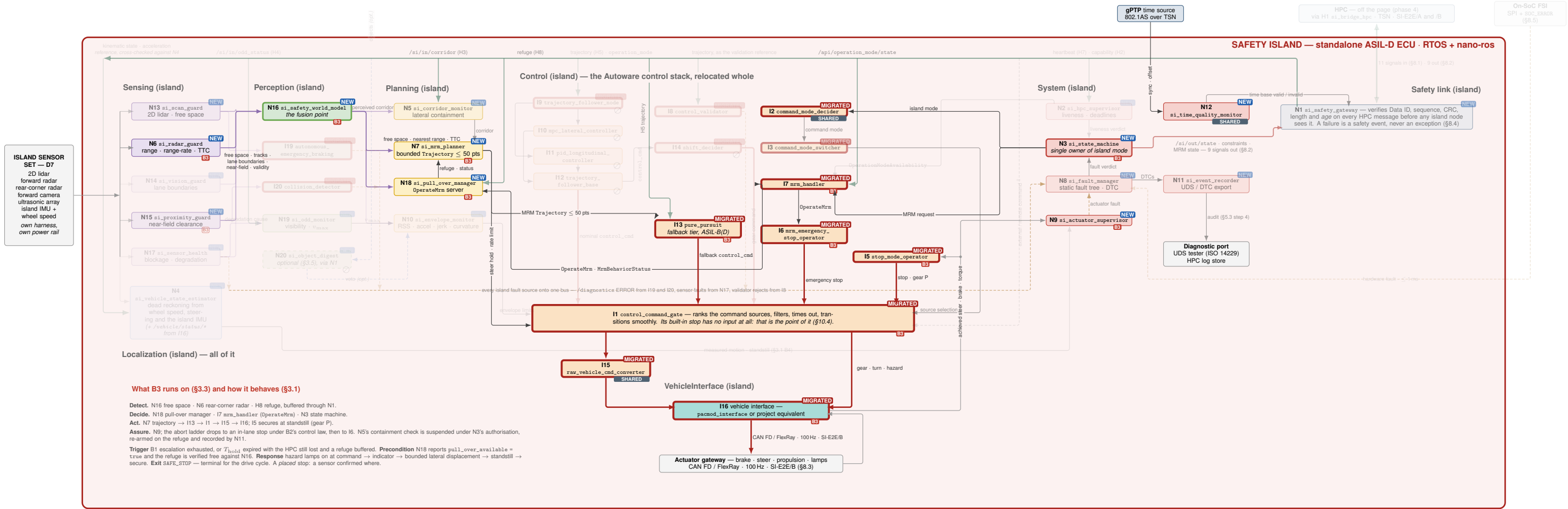
In this view — B2, lane-keep hold



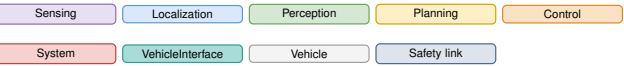
Every edge on this page, with the design-document section it is taken from, is catalogued in ../messageflow/, which draws the same 37 nodes coloured by execution partition instead.

Phase 5 · B3 — Pull over if the ODD continues to fail

When degradation does not restore the envelope within a declared deadline, bring the vehicle out of the running lane to a standstill in a nominated refuge.
The phase-5 drawing, unchanged: what B3 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



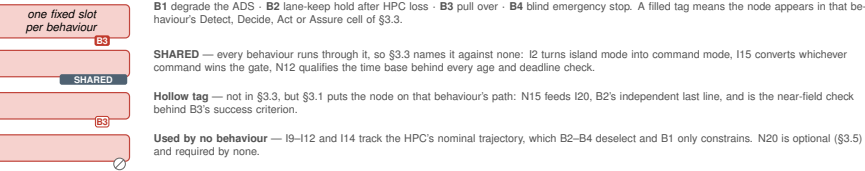
Stacks — the colours of ArchitectureMain.pdf



Where each node came from



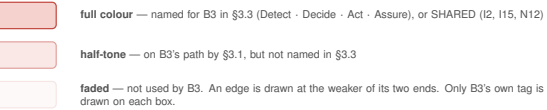
Behaviours each node enables (§3.3)



Messages



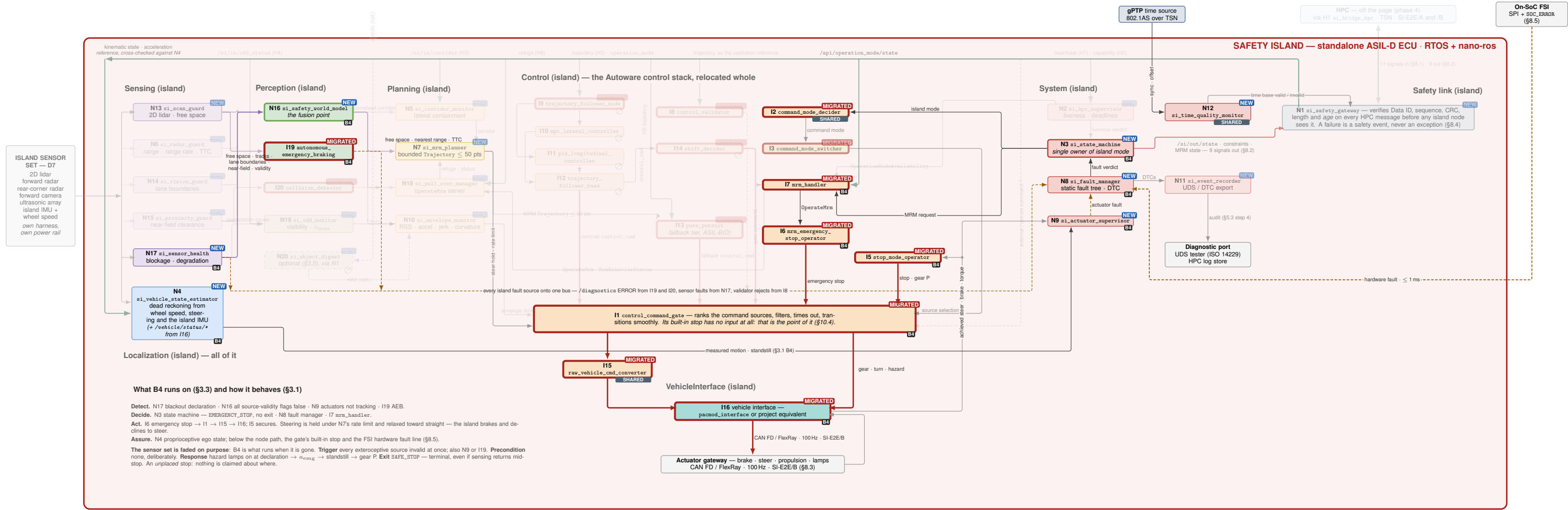
In this view — B3, pull over



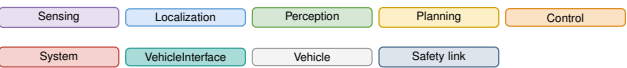
Every edge on this page, with the design-document section it is taken from, is catalogued in ../messageflow/, which draws the same 37 nodes coloured by execution partition instead.

Phase 5 · B4 — Emergency stop when the island is blind

When the island has lost the sensing that B2 and B3 both depend on, decelerate to a standstill where the vehicle already is — no lane keeping, no refuge, no lateral manoeuvre. The phase-5 drawing, unchanged: what B4 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



Stacks — the colours of ArchitectureMain.pdf

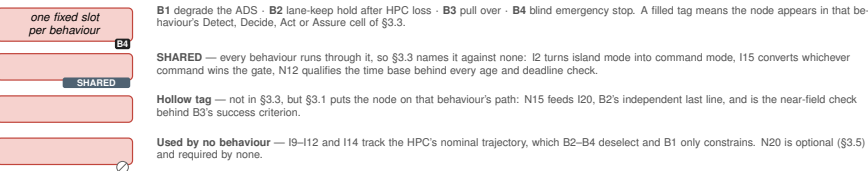


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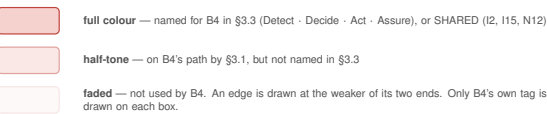
Behaviours each node enables (§3.3)



Messages



In this view — B4, blind emergency stop



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